



MIKROSKIL IDEAFUSE COMPETITION 2025

Complete Walk

You are given a $2 \times N$ grid, where some cells may be blocked. The rows are numbered from 1 to 2, and the columns are numbered from 1 to N . You can move from a cell (r, c) to any of its adjacent cells, i.e. $(r + 1, c)$, $(r - 1, c)$, $(r, c + 1)$, or $(r, c - 1)$, as long as the move stays within the grid and the destination cell is not blocked.

A walk is a sequence of one or more adjacent, unblocked cells. A walk is considered complete if and only if it visits every unblocked cell exactly once (no repeats).

Your task is to count the number of unique complete walks possible on the given grid, output the result modulo 1,000,000,007.

Two complete walks A and B are considered different if there exists at least one position i (from 1 to $2N - M$, where M is the number of blocked cells) such that the i -th cell in walk A is not the same as in walk B.

For example, let $N = 4$, and there is $M = 1$ blocked cell at position $(1, 2)$.

The four possible complete walks are:

- $(1, 1), (2, 1), (2, 2), (2, 3), (2, 4), (1, 4), (1, 3)$
- $(1, 1), (2, 1), (2, 2), (2, 3), (1, 3), (1, 4), (2, 4)$
- $(1, 3), (1, 4), (2, 4), (2, 3), (2, 2), (2, 1), (1, 1)$
- $(2, 4), (1, 4), (1, 3), (2, 3), (2, 2), (2, 1), (1, 1)$

Input Format

The input starts with an integer: T ($1 \leq T \leq 1000$) representing the number of cases.

Each case begins with two integers: N and M ($1 \leq N \leq 10^9$; $0 \leq M \leq 2N$), where N is the grid size, and M is the number of blocked cells in the grid. The next M lines each contains two integers: R_i and C_i ($1 \leq R_i \leq 2$; $1 \leq C_i \leq N$) representing a blocked cell (R_i, C_i) in the grid. The given blocked cells are guaranteed to be unique.

The total sum of M across all cases does not exceed 500,000.

Output Format

For each case, output a line in the format: "Case #X: Y", where X is the case number (starting from 1) and Y is the output for that case, modulo 1,000,000,007.

Sample Input

```
4
4 1
1 2
3 0
5 2
1 2
2 3
2 4
1 1
1 2
2 1
2 2
```

Sample Output

```
Case #1: 4
Case #2: 16
Case #3: 0
Case #4: 1
```

Explanation Of Sample Input/Output

In case #4, as all the cells are blocked, the only valid complete walk contains no cell.